



NAME – RISHABH ANAND

ROLL NO. – 20256709

PROJECT REPORT – CAR ACCIDENT PREVENTION USING ARDUINO IR SENSORS

GROUP NO - 18

Car Accident Prevention System Using IR Sensors

Introduction

Road accidents at sharp curves, blind junctions, and narrow crossroads are extremely common, especially in areas where visibility is low and drivers cannot see incoming vehicles from the perpendicular road. Such situations often lead to severe collisions simply because drivers do not receive timely warnings about approaching vehicles. To address this issue, this project presents a low-cost, Arduino-based accident prevention system that uses Infrared (IR) sensors to detect vehicles in real time and warn drivers using LED indicators and an audio buzzer alert.

The fundamental idea behind this project is to create an automated signalling system for two roads intersecting at a blind junction. When a vehicle approaches from either side, the system instantly detects it and displays an appropriate signal: green light for the road that is allowed to move and red light for the road that must stop. When both vehicles approach simultaneously and at similar speeds, the system forces both to stop by turning both LEDs red and activating the buzzer. This helps prevent panic braking, sudden collisions, and confusion at blind turns.

Objective

The primary objective of this project is to design a simple, reliable, and cost-effective accident prevention mechanism for blind spots using IR-based detection. The system is intended to:

- Detect vehicles approaching from two different roads.
- Automatically assign right-of-way using LED indicators.
- Warn drivers through a buzzer when both vehicles approach together.
- Reduce the likelihood of collision by providing early and clear warnings.
- Demonstrate a microcontroller-based safety prototype suitable for real-world adaptation.

Components Used

The system is built using basic electronic components, all easily available and inexpensive:

- Arduino Uno / Nano – Acts as the central microcontroller controlling the logic.
- IR Sensors (2 units) – Detect incoming vehicles by sensing interruptions in IR light.

- Red and Green LEDs (4 total) – Provide visible stop/go signals to both sides.
- Buzzer – Generates sound alerts when both vehicles approach.
- 220Ω resistors – Used to limit current for LEDs.
- Breadboard and jumper wires – For connections.
- 5V USB/Power Supply – Powers the Arduino.

Working Principle

The system functions by continuously monitoring both IR sensors placed on the two approaching roads. Each sensor gives a digital output that indicates whether a car is present.

Case 1: Vehicle only on Road 1 (Car 1)

The system gives right-of-way to Road 1. The LED on Road 1 glows green, while Road 2 gets a red light. The buzzer remains off because no collision risk exists.

Case 2: Vehicle only on Road 2 (Car 2)

The logic is reversed: Road 2 gets a green signal and Road 1 gets a red light, again with no buzzer.

Case 3: Vehicles on both roads at the same time

This is the most critical scenario. When both IR sensors detect vehicles simultaneously, it means both cars are approaching the blind junction together. In this case:

- Both roads are shown red lights, instructing drivers to slow down.
- The buzzer starts buzzing, creating an immediate alert.

This ensures both drivers pause and avoid rushing into the junction blindly.

Circuit Description

The IR sensors are connected to digital pins 2 and 3 of the Arduino. LEDs for both roads are connected to pins 4, 5, 6, and 7, while the buzzer is connected to pin 8. All sensors and LEDs share a common ground. The circuit is simple enough to be assembled on a breadboard but can also be soldered into a PCB for permanent use.

Arduino Code

```
// Accident Prevention System for Crossroads
```

```
// Using Arduino Nano, 2 IR sensors, 4 LEDs, and a buzzer
```

```
int ir1 = 2; // IR Sensor 1 (Road 1)
int ir2 = 3; // IR Sensor 2 (Road 2)
int red1 = 4; // Road 1 Red LED
int green1 = 5; // Road 1 Green LED
int red2 = 6; // Road 2 Red LED
int green2 = 7; // Road 2 Green LED
int buzzer = 8; // Buzzer

void setup()
{

    pinMode(ir1, INPUT);
    pinMode(ir2, INPUT);
    pinMode(red1, OUTPUT);
    pinMode(green1, OUTPUT);
    pinMode(red2, OUTPUT);
    pinMode(green2, OUTPUT);
    pinMode(buzzer, OUTPUT);

    // Initialize all OFF
    digitalWrite(red1, LOW);
    digitalWrite(green1, LOW);
    digitalWrite(red2, LOW);
    digitalWrite(green2, LOW);
    digitalWrite(buzzer, LOW);
}

void loop()
{
    int s1 = digitalRead(ir1); // Read IR1 state
```

```
int s2 = digitalRead(ir2); // Read IR2 state

// IR sensors usually give LOW when an object (car) is detected
bool car1 = (s1 == LOW);
bool car2 = (s2 == LOW);

// CASE 1: Cars on both roads → possible accident
if (car1 && car2)
{
    digitalWrite(red1, HIGH);
    digitalWrite(green1, LOW);
    digitalWrite(red2, HIGH);
    digitalWrite(green2, LOW);
    digitalWrite(buzzer, HIGH); // Sound alarm
}

// CASE 2: Car only on Road 1
else if (car1 && !car2)
{
    digitalWrite(green1, HIGH); // Road 1 GO
    digitalWrite(red1, LOW);
    digitalWrite(red2, HIGH); // Road 2 STOP
    digitalWrite(green2, LOW);
    digitalWrite(buzzer, LOW);
}

// CASE 3: Car only on Road 2
else if (car2 && !car1)
{
    digitalWrite(green2, HIGH); // Road 2 GO
    digitalWrite(red2, LOW);
    digitalWrite(red1, HIGH); // Road 1 STOP
    digitalWrite(green1, LOW);
    digitalWrite(buzzer, LOW);
}
```

```
digitalWrite(red1, HIGH); // Road 1 STOP
digitalWrite(green1, LOW);
digitalWrite(green2, HIGH); // Road 2 GO
digitalWrite(red2, LOW);
digitalWrite(buzzer, LOW);
}
```

```
// CASE 4: No car detected → both GREEN
```

```
Else
```

```
{
    digitalWrite(red1, LOW);
    digitalWrite(green1, HIGH);
    digitalWrite(red2, LOW);
    digitalWrite(green2, HIGH);
    digitalWrite(buzzer, LOW);
}
```

```
delay(200); // small delay for stable readings
```

```
}
```

Applications

This project can be used in several real-world situations such as:

- Blind curves and turns on hilly or rural roads.
- Intersections without traffic signals.
- Dense urban lanes where visibility is often blocked.
- School zones, hospital stretches, and pedestrian crossings.
- Private roads and campus routes.

Advantages

- Very low cost and easy to implement.

- Works in real time with minimal delay.
- Requires no complicated hardware.
- Reduces collision chances significantly.
- Perfect for areas where installing full traffic signals is impractical.

Limitations

- IR sensors may face reduced accuracy in harsh weather.
- Range is limited depending on sensor quality.
- Requires stable power and careful alignment.

Future Scope

The system can be enhanced further by:

- Replacing IR sensors with ultrasonic or LIDAR sensors.
- Adding wireless communication between vehicles.
- Integrating a display board showing traffic status.
- Using solar panels for power.
- Adding machine learning to predict traffic patterns.

Conclusion

This project demonstrates a functional and efficient prototype that can significantly reduce accidents at blind junctions using budget-friendly hardware. By combining IR detection with clear visual and audio alerts, the system improves driver awareness and enhances road safety. With further development, this design can be adapted into a practical real-world solution used in rural and urban areas.